1 Executive summary

One frozen Perovskite 0292 acquisition | 20% Hadamard | NPL-default noise

Question. Does replacing LION's interpolation-style primal relaxation with canonical Chambolle-Pock extrapolation improve reconstruction or optimisation?

Answer. For TV, TGV-2 and TGV-3, the reconstructed images are almost identical after 8,192 iterations. The canonical update gives a lower variational objective, often substantially earlier.

Largest benefit. The objective falls much faster for TGV-2 and TGV-3. Canonical CP at 512 iterations already beats the legacy objective at 8,192 iterations for both models.

Practical limit. Faster objective convergence does not imply better PSNR or preserved details on this one case. Each PDHG iteration has nearly the same cost under either rule.

Quality and objective after 8,192 iterations

Method	Relaxation	PSNR (dB)	SSIM	HaarPSI	Objective	Time (s)
TV	LION interpolation	30.369715	0.814062	0.725788	47.61201	7.846
TV	Canonical CP	30.369713	0.814064	0.725789	47.60939	7.889
TGV-2	LION interpolation	30.436258	0.740215	0.734538	28.24963	12.174
TGV-2	Canonical CP	30.436237	0.740215	0.734535	27.84567	12.316
TGV-3	LION interpolation	30.464329	0.740222	0.733572	29.11536	16.640
TGV-3	Canonical CP	30.464373	0.740227	0.733584	27.87157	16.782

Effect size and per-iteration computational cost

Method	Objective reduction	Delta PSNR (dB)	LION ms/step	CP ms/step
TV	0.0055%	-0.0000019	0.958	0.963
TGV-2	1.4300%	-0.0000210	1.486	1.503
TGV-3	4.2719%	+0.0000439	2.031	2.049

Single-image oracle scalar weights are held fixed. All metrics are on unmodified reconstructions.

2 Experiment design and controlled variables

Same algorithms, identical parameters; only the relaxed-primal update differs

Existing LION PDHG uses interpolation: zbar[n+1] = (1-rho) z[n] + rho z[n+1]. Here rho=1, so zbar[n+1] = z[n+1].

Canonical Chambolle-Pock uses extrapolation: zbar[n+1] = z[n+1] + theta (z[n+1]-z[n]). Here theta=1, so zbar[n+1] = 2 z[n+1]-z[n].

For all three methods, the solver's initial primal and dual variables were zero; lambda/alpha weights, sampling indices, measured currents, normalisation, boundary conditions, and step sizes (tau=sigma=0.18) were unchanged. We used two local solver copies differing by one line; no LION source was modified.

PSNR-optimised scalar weights, selected previously on this image

Method	Fixed scalar weights	PDHG iterations
TV	lambda = 0.0240982484	8,192
TGV-2	alpha1 = 0.0282842712, alpha0 = 0.01	8,192
TGV-3	alpha2 = 0.032, alpha1 = 0.01, alpha0 = 0.024	8,192

Convergence recording and validation

- Nine independent zero-start runs per rule: iterations 1, 2, 64, 256, 512, 1,024, 2,048, 4,096 and 8,192. The displayed final full reconstructions and metric tables use 8,192 iterations.

- The copied legacy implementation reproduced each original archived TV, TGV-2 and TGV-3 reconstruction exactly, pixel for pixel. The two rules agree after the first iteration and differ from the second.

- Objective is the complete method-specific primal functional: one half of the squared normalized PCM data residual plus the method's fixed scalar TV/TGV penalty. Objective values are comparable between relaxation rules for the SAME method, but not across TV, TGV-2 and TGV-3.

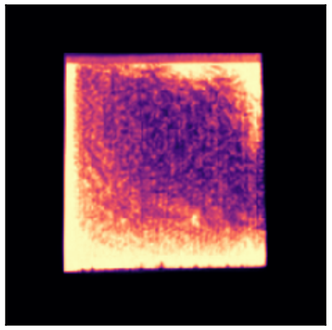
- Runtime is the elapsed time for each independent GPU run, including the solver loop and its per-run startup. Per-iteration runtime is total runtime divided by iteration count, so it is a useful estimate rather than an isolated kernel benchmark.

Provenance: original PCM frozen test data; local LION/MRPro PDHG copies; NPL-default noise.

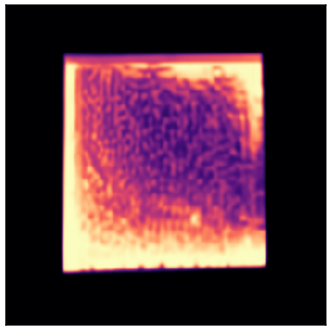
3 Full-image reconstruction comparison

GT, PnP-ADMM and UTV benchmarks plus the six relaxation-rule reconstructions

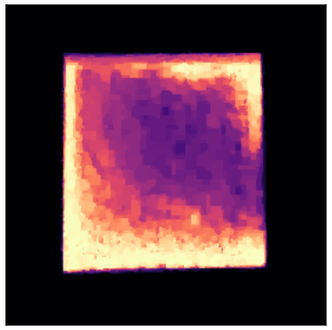
Ground truth



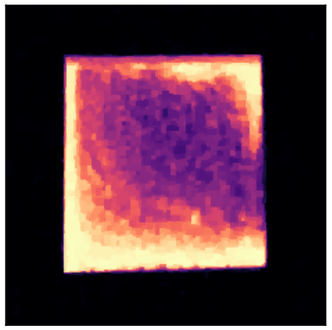
PnP-ADMM-tuned-5
PSNR 32.369354 dB



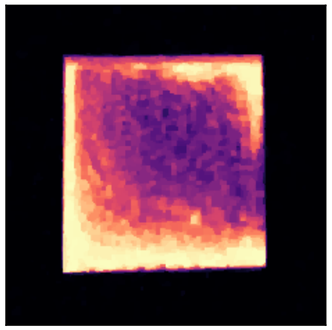
UTV C60, best rho
PSNR 29.408306 dB



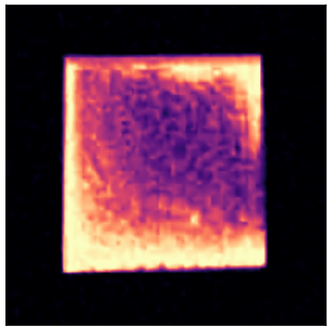
TV | LION
PSNR 30.369715 dB | SSIM 0.814062
HaarPSI 0.725788 | 8,192 steps



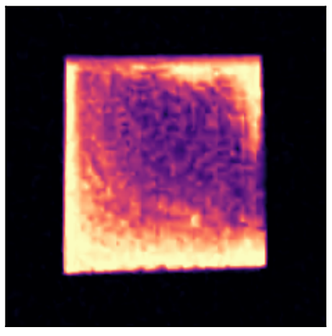
TV | Canonical CP
PSNR 30.369713 dB | SSIM 0.814064
HaarPSI 0.725789 | 8,192 steps



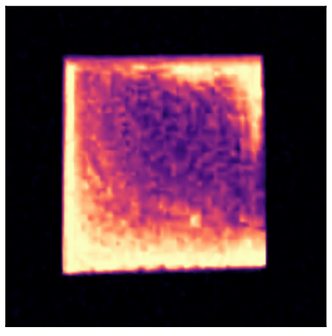
TGV-2 | LION
PSNR 30.436258 dB | SSIM 0.740215
HaarPSI 0.734538 | 8,192 steps



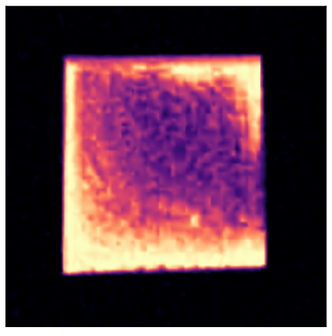
TGV-2 | Canonical CP
PSNR 30.436237 dB | SSIM 0.740215
HaarPSI 0.734535 | 8,192 steps



TGV-3 | LION
PSNR 30.464329 dB | SSIM 0.740222
HaarPSI 0.733572 | 8,192 steps



TGV-3 | Canonical CP
PSNR 30.464373 dB | SSIM 0.740227
HaarPSI 0.733584 | 8,192 steps



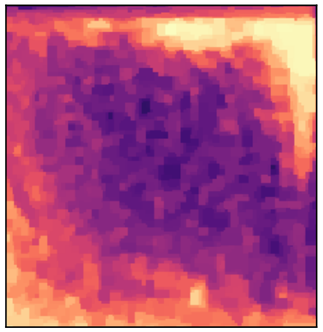
All panels use the same fixed [0,1] intensity range. Metrics use the original raw (unclipped) reconstructions.

The canonical images are ACTUAL reconstructed arrays, not placeholders or copies of the legacy panels.

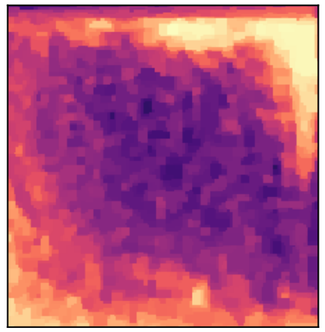
4 Matched texture crops and signed differences

Each row compares the same method under both relaxation rules

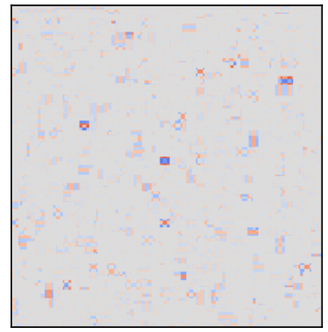
TV | LION
PSNR 30.369715 dB



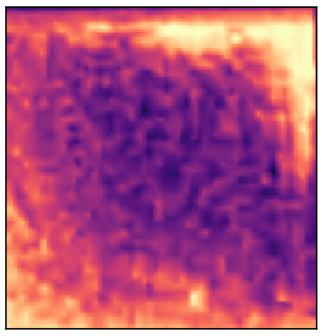
TV | Canonical CP
PSNR 30.369713 dB



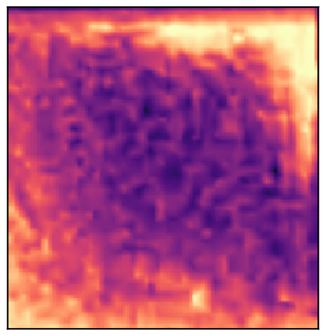
TV | CP - LION
Max |difference|: 3.263e-05



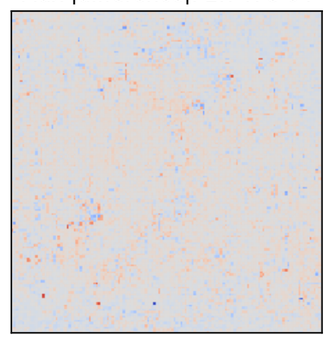
TGV-2 | LION
PSNR 30.436258 dB



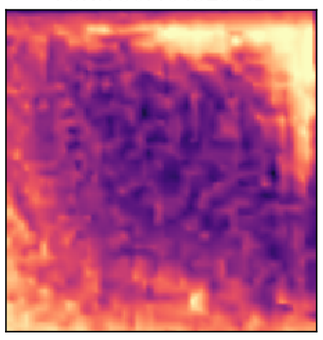
TGV-2 | Canonical CP
PSNR 30.436237 dB



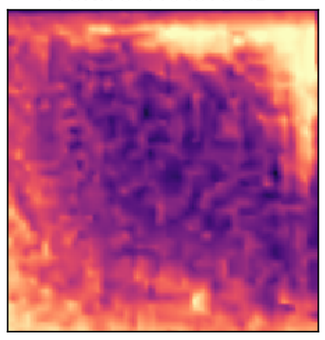
TGV-2 | CP - LION
Max |difference|: 2.747e-04



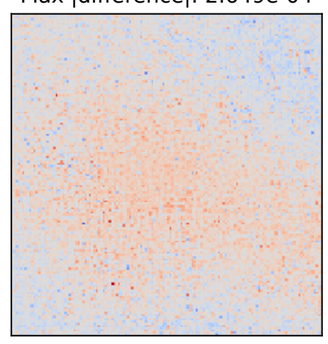
TGV-3 | LION
PSNR 30.464329 dB



TGV-3 | Canonical CP
PSNR 30.464373 dB



TGV-3 | CP - LION
Max |difference|: 2.649e-04

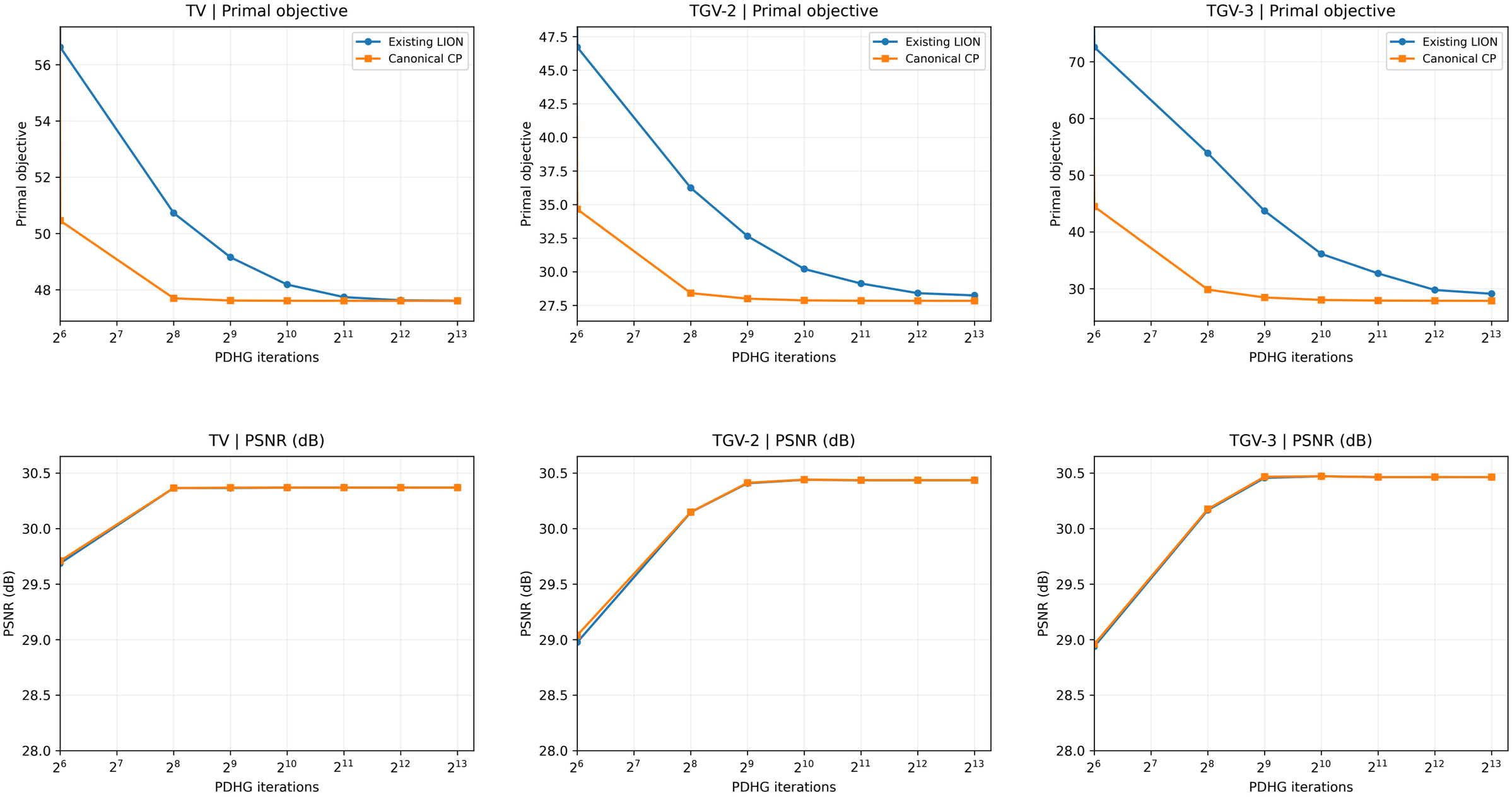


Identical crop: rows 40-184, columns 65-204. Signed differences use a separate symmetric scale for each method.

Small visual differences should not be mistaken for equivalence of the auxiliary TGV fields.

5 Objective convergence and reconstruction quality

Upper row: method-specific objectives. Lower row: PSNR at the same iteration checkpoints



The curves show independently restarted runs. Canonical CP reduces the objective much sooner, especially for TGV-2 and TGV-3.

The converged image quality need not improve even when the full variational objective decreases faster.

6 Iteration cost, savings and interpretation

Objective improvements at a fixed iteration budget versus a smaller computational budget

Method	LION 8192 s	CP 8192 s	LION ms/step	CP ms/step	CP objective @512	LION objective @8192	CP 512 s
TV	7.846	7.889	0.958	0.963	47.62022	47.61201	0.499
TGV-2	12.174	12.316	1.486	1.503	28.00785	28.24963	0.773
TGV-3	16.640	16.782	2.031	2.049	28.46180	29.11536	1.045

- Time per iteration. At 8,192 iterations, TV requires about 0.958 ms/step, TGV-2 about 1.486 ms/step, and TGV-3 about 2.031 ms/step under the existing rule. The canonical update adds almost no per-step cost.

- Relative cost. On this GPU, the TGV-2 step takes about 1.55x the TV step, while TGV-3 takes about 2.12x the TV step and 1.37x the TGV-2 step.

- Objective threshold benefit. Canonical TGV-2 and TGV-3 each reach a lower objective after 512 iterations than existing LION after 8,192. For TV, canonical CP needs 1,024 iterations to reach a lower objective than LION after 8,192. This does not certify a global optimum.

- Image quality caveat. Final PSNR changes are below 0.00005 dB in all three models. Earlier stopping under canonical CP may reduce compute time if the desired criterion is objective convergence, but should not be used without checking image quality and optimality gaps.

- Cross-method caution. The three models have different regularisers and objectives. Compare LION versus CP objectives within one method only. Compare TV/TGV-2/TGV-3 using reconstruction quality or wall-clock time, not their absolute objective magnitudes.

Primary references

Chambolle and Pock (2011), DOI 10.1007/s10851-010-0251-1; Bredies, Kunisch and Pock (2010), DOI 10.1137/090769521.